

Taesoo Kim

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SUMMARY

Signal and power integrity researcher specializing in advanced packaging and heterogeneous integration (HBM, chiplets, glass-core interposers). 13 IEEE conference publications (4 first-author) with Best Paper and Best Poster awards; SK hynix Global Ph.D. Fellow; industry SI/PI experience at Cisco and Samsung.

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Ph.D. in Electrical and Computer Engineering, Aug 2025 – present (expected Aug 2030)

Advisor: Prof. Muhannad S. Bakir | GPA: 4.0/4.0

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, Korea

M.S. in Electrical Engineering, Aug 2025

Advisor: Prof. Joungho Kim (Terabyte Interconnection and Package Laboratory) | GPA: 3.80/4.0

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, Korea

B.S. in Electrical Engineering, Feb 2023

Magna Cum Laude; selected for the Young Engineer Honor Society (YEHS), National Academy of Engineering of Korea, by recommendation of the Dean of the College of Engineering

EXPERIENCE

Cisco Systems, Inc. – Signal Integrity / Hardware Engineering Intern

San Jose, CA | Jun – Aug 2026

- Developed a geometry-aware Python solver that predicts PCB fiber-weave-induced differential skew, combining Monte Carlo multi-ply glass-placement statistics with frequency-dependent coupled RLGC analysis, mixed-mode S-parameters, and group-delay skew extraction over arbitrary 2D routing.
- Correlated the solver against HFSS/Touchstone reference simulations, analytical baselines, and a legacy MATLAB model; delivered it as a reproducible internal tool with 49 automated regression tests and maintainer documentation.
- Performed PCIe signal integrity analysis and lab support: Cadence Allegro net and test-point correlation, TDR measurement interpretation, and high-speed interface test planning.

Samsung Semiconductor, Inc. – SI/PI Engineering Intern

San Jose, CA | Jun – Aug 2024

- Performed SI/PI analysis and optimization of UCIe-A channels for high-speed chiplet interconnects, contributing to a successful chip tape-out.
- Conducted full-wave EM simulations to minimize crosstalk and insertion loss, ensuring robust eye margin for next-generation links.

KAIST Terabyte Interconnection and Package Laboratory – Graduate Research Assistant

Daejeon, Korea | 2023 – 2025

- Led the design of next-generation HBM architectures: Twin Tower HBM (**Best Paper Award, IEEE EDAPS 2024**) and Quad Tower HBM on glass substrates with local silicon bridges (**Best Poster Award, IEEE EPEPS 2025**).
- Designed and verified a UCIe-A mimic channel enabling early on-chip system validation of UCIe-based designs (IEEE ECTC 2025).
- Managed two industry-funded research projects on SI/PI modeling for 2.5D/3D heterogeneous integration.

KAIST Computer, Architecture, System and Technology Lab – Undergraduate Research Assistant

Daejeon, Korea | 2022

- Designed high-efficiency DMA controller architectures, optimizing data transfer between SRAM and logic units for AI accelerators.

SELECTED PUBLICATIONS (FIRST AUTHOR)

- [1] **T. Kim**, S. Penta, S. Woo, and M. S. Bakir, "A Design-Space Exploration of Active Interposer Architectures for Long-Reach Signaling in Large-Area Advanced Packages," *Proc. IEEE 76th Electronic Components and Technology Conference (ECTC)*, 2026, pp. 1952–1956.
- [2] **T. Kim**, Y. Yoon, H. Suh, H. An, J. Yoon, J. Park, J. Bae, B. Kim, I. Choi, E. Seo, J. Koo, and J. Kim, "Quad Tower High Bandwidth Memory (QT-HBM) with Glass Substrate and Local Silicon Bridge," *Proc. IEEE 34th Conference on Electrical Performance of Electronic Packaging and Systems (EPEPS)*, 2025. **Best Poster Award.**
- [3] **T. Kim**, T. Shin, K. Kim, H. Suh, H. An, and J. Kim, "UCle-A Mimic Channel Design and Verification for Early On-Chip System Validation," *Proc. IEEE 75th Electronic Components and Technology Conference (ECTC)*, 2025, pp. 2199–2205.
- [4] **T. Kim**, J. Yoon, S. Choi, H. Kim, H. Suh, H. An, J. Ahn, H. Park, and J. Kim, "Design and Analysis of Twin Tower High Bandwidth Memory (HBM) Architecture for Large Memory Capacity and High Bandwidth System," *Proc. IEEE Electrical Design of Advanced Packaging and Systems (EDAPS)*, 2024. **Best Paper Award.**

9 additional co-authored IEEE publications (ECTC, EPEPS, 2023 – 2025) on PDN optimization, ML-assisted SI/PI design, and GPU–HBM architectures. Full list: taesoo-k.github.io/publications

HONORS & AWARDS

- SK hynix Global Ph.D. Fellowship, SK hynix Inc. (2026)
- Best Poster Award, IEEE Conference on Electrical Performance of Electronic Packaging and Systems (EPEPS) (2025)
- Best Paper Award, IEEE Electrical Design of Advanced Packaging and Systems (EDAPS) (2024)
- National Science and Technology Scholarship, Republic of Korea

SKILLS

- **SI/PI & EM simulation:** Ansys HFSS, SIwave, Q3D, Maxwell; Keysight ADS; Cadence Allegro, Virtuoso
- **Measurement & validation:** TDR analysis, simulation–measurement correlation, S-parameter/Touchstone workflows
- **Programming:** Python (NumPy, SciPy, scikit-learn, Matplotlib), MATLAB, Verilog; automated testing, reproducible pipelines
- **ML for design:** reinforcement learning and generative methods for interconnect/PDN optimization

LEADERSHIP & SERVICE

- President, Georgia Tech ECE Korean Student Association, Atlanta, GA (2026 – present)
- Vice President, Korean Student Association, Georgia Tech, Atlanta, GA (2025)
- President, Graduate Student Association of Electrical Engineering, KAIST (2024)
- Sergeant, Republic of Korea Army – Detection Analysis Specialist, ROK Defense Ground Operations Command (2020 – 2021)